

"Battery for Better Life"

POWER EVOLUTION

# ***SUPREME***

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**SEBANG**

SEBANG GLOBAL BATTERY



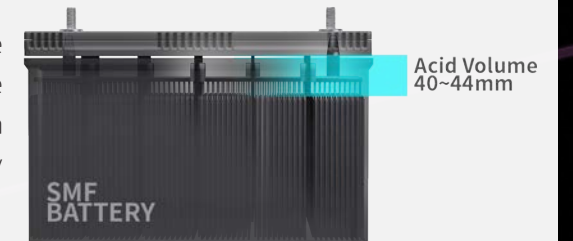
# POWER EVOLUTION SUPREME

## Battery for better life

Sebang Global Battery is endeavoring for a better future.  
By producing various brands and type of batteries with the best technology,  
we are demonstrating exceptional innovation around the world.

### Design Features

Designed to keep a proper level of electrolyte in the upper part of plates via accommodating an acid volume by 10nm or higher than competitors' to sustain an acid volume in case of water consumption caused by evaporation at a high temperature.

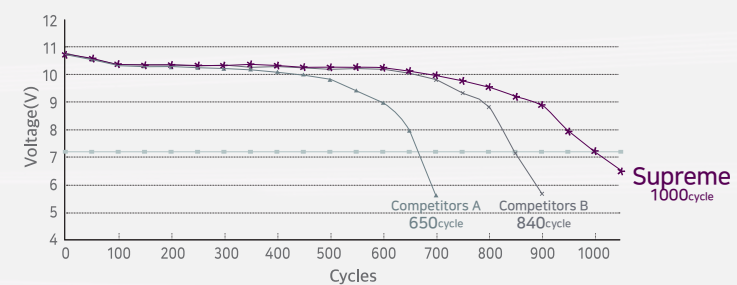


### Application of cold-forged bushing & leakage-proof



### An increase of life span via adoption of spun-bonded paper

- An increase of life span via adoption of spun-bonded paper
- An increase of life span via maintenance of active materials of positive plates

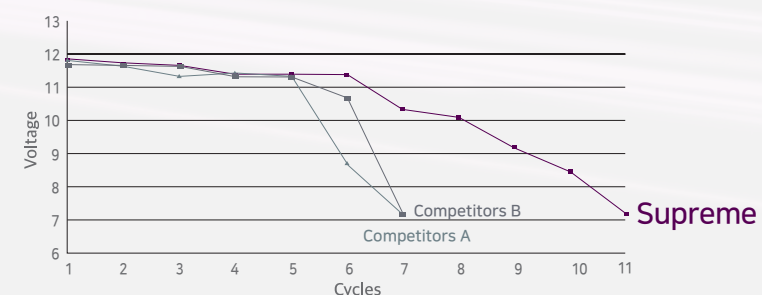


### An increment of anti-corrosion & life span via an adoption of grid in high Sn(tin)

#### Ranking in life span under overcharging of high temperature

Ranking in an anti-corrosion under the condition of specific gravity 1,280 at 25 degrees celsius of an ambient temperature

1. High Sn
2. Low Sn
3. Cast



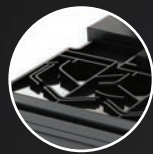
# POWER EVOLUTION SUPREME

## ① Hydrometer

- At full charge, the electrolyte specific gravity is 1.28 while at 50% of charge considered the minimum serviceable condition, the specific gravity is typically 1.22, in a typical situation. When the specific gravity drops to 1.10. The battery is full discharged.

## ② Double lid with labyrinth structure

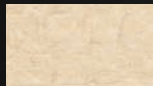
- Prevent electrolyte losses by collection and returning liquid to the reservoir.
- Vents allow the battery to “breathe” during temperature changes and charging.



## ③ Flame Arrester

- Safety system
- Prevent the explosion from spark of outside
- Minimize the acid-leakage
- Prevent the inflow of dust

### NST Special Tissue for Positive Plates



Prevent the active material shedding from frequent charging and discharging.

### REG Rolled Expanded High Tin Grid



Creating sub grain boundary which provide longer life span against corrosion. Ergonomic easy carry down fold handle

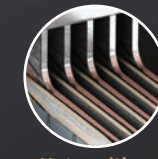
## ④ Wrought lead-Calcium Grids

- Offer considerable strength
- Resistant to grid corrosion
- Over-charge resistant
- Minimal gassing and water usage
- Less self-discharge
- Resist thermal runaway

Diamond Mesh – increase the conductivity  
Central lug – increase the electricity collection characteristic  
High density active material with special additives – high starting performance



Diamond Mesh



Hot-melting

Improve the anti-vibration characteristic

## ⑤ Double lid

- Prevent contamination and also add to the strength and rigidity.
- Include permanent flame arrester to prevent accidental explosion from external sparks or flame.
- Have hydrometer built in for faster checking.

Cold forging terminal - High strength from fine structure and prevent the leakage



## ⑥ Centered Cast-on Plate Straps

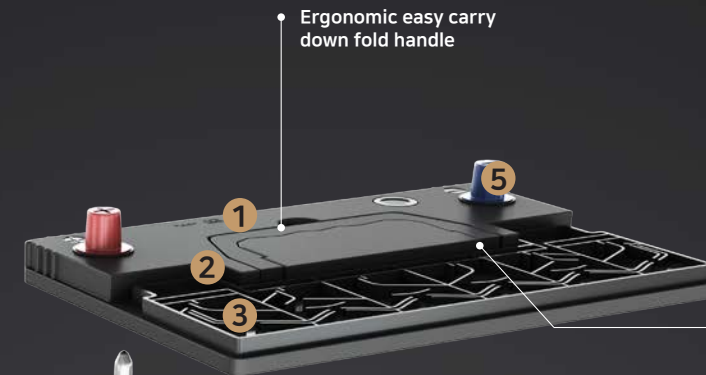
- Stronger than the thinner gas-burned conventional connectors.
- Reduce the lever action movement resulting from road shock.

## ⑦ Polypropylene Case

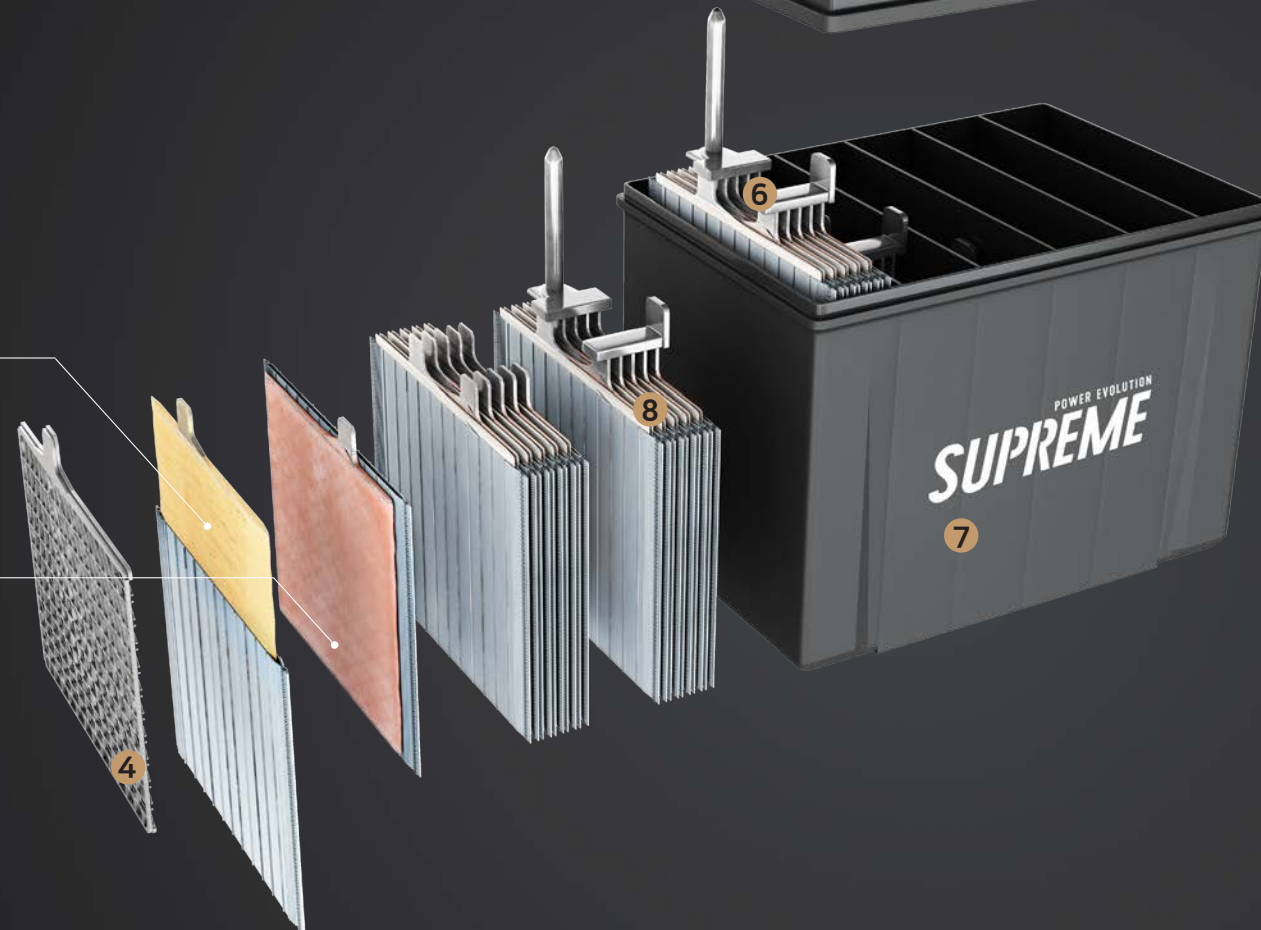
- Reinforced design is precisely tailored to support the battery elements.
- Material is exceptionally strong, durable and withstands the shock and vibration from the road.

## ⑧ Low-Resistance Envelope Separators

- Encapsulate negative plates
- Improve vibration durability
- Prevent “treeing” and internal shorting between positive and negative plates.



Ergonomic easy carry down fold handle



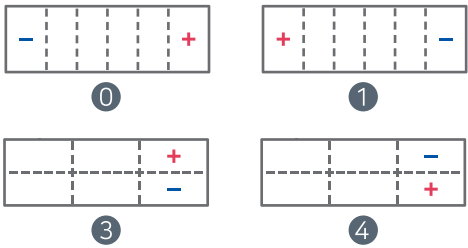


JIS Type

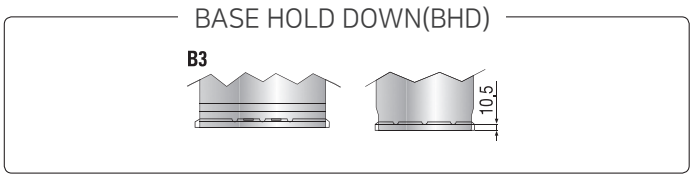
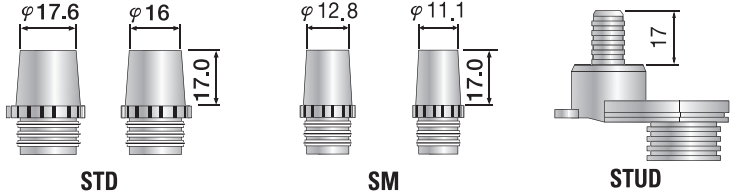
JAPANESE INDUSTRIAL STANDARDS TYPE CALCIUM MF BATTERY



CELL LAYOUT



TERMINAL



Sealed Maintenance Free Battery(ASIA)

Passenger Car & Light Commercial Battery

| Old JIS        | New JIS     | Dimensions(mm) |     |     | Voltage (V) | Capacity(Ah) |      | RC (Min) | CCA (-18℃) | CA (0℃) | Layout | Terminal | Handle |
|----------------|-------------|----------------|-----|-----|-------------|--------------|------|----------|------------|---------|--------|----------|--------|
|                |             | L              | W   | TH  |             | 5HR          | 20HR |          |            |         |        |          |        |
| B17            | SMF 34B17L  | 167            | 127 | 227 | 12          | 27           | 33   | 47       | 280        | 390     | 0      | SM       | P/H    |
|                | SMF 34B17R  | 167            | 127 | 227 | 12          | 27           | 33   | 47       | 280        | 390     | 1      | SM       | P/H    |
| SMF NS40ZL     | SMF 40B19L  | 187            | 127 | 226 | 12          | 27           | 35   | 49       | 300        | 410     | 0      | SM       | P/H    |
| SMF NS40Z      | SMF 40B19R  | 187            | 127 | 226 | 12          | 27           | 35   | 49       | 300        | 410     | 1      | SM       | P/H    |
| SMF NS40ZLS    | SMF 40B19LS | 187            | 127 | 226 | 12          | 27           | 35   | 49       | 300        | 410     | 0      | STD      | P/H    |
| SMF NS40ZS     | SMF 40B19RS | 187            | 127 | 226 | 12          | 27           | 35   | 49       | 300        | 410     | 1      | STD      | P/H    |
| SMF NX100-S6L  | SMF 55B24L  | 238            | 129 | 225 | 12          | 36           | 45   | 80       | 430        | 540     | 0      | SM       | P/H    |
| SMF NX100-S6   | SMF 55B24R  | 238            | 129 | 225 | 12          | 36           | 45   | 80       | 430        | 540     | 1      | SM       | P/H    |
| SMF NX100-S6LS | SMF 55B24LS | 238            | 129 | 225 | 12          | 36           | 45   | 80       | 430        | 540     | 0      | STD      | P/H    |
| SMF NX100-S6S  | SMF 55B24RS | 238            | 129 | 225 | 12          | 36           | 45   | 80       | 430        | 540     | 1      | STD      | P/H    |
| SMF 50D20L     | SMF 50D20L  | 202            | 173 | 225 | 12          | 42           | 50   | 85       | 450        | 590     | 0      | STD      | P/H    |
| SMF 50D20R     | SMF 50D20R  | 202            | 173 | 225 | 12          | 42           | 50   | 85       | 450        | 590     | 1      | STD      | P/H    |
| D23            | SMF 55D23L  | 232            | 173 | 225 | 12          | 48           | 60   | 100      | 500        | 630     | 0      | STD      | P/H    |
|                | SMF 55D23R  | 232            | 173 | 225 | 12          | 48           | 60   | 100      | 500        | 630     | 1      | STD      | P/H    |
|                | SMF 75D23L  | 232            | 173 | 225 | 12          | 52           | 65   | 110      | 580        | 710     | 0      | STD      | P/H    |
|                | SMF 75D23R  | 232            | 173 | 225 | 12          | 52           | 65   | 110      | 580        | 710     | 1      | STD      | P/H    |
| SMF N50L       | SMF 48D26L  | 260            | 173 | 222 | 12          | 40           | 50   | 80       | 420        | 550     | 0      | STD      | P/H    |
| SMF N50        | SMF 48D26R  | 260            | 173 | 222 | 12          | 40           | 50   | 80       | 420        | 550     | 1      | STD      | P/H    |
| SMF N50ZL      | SMF 55D26L  | 260            | 173 | 222 | 12          | 48           | 60   | 100      | 500        | 630     | 0      | STD      | P/H    |
| SMF N50Z       | SMF 55D26R  | 260            | 173 | 222 | 12          | 48           | 60   | 100      | 500        | 630     | 1      | STD      | P/H    |
| SMF NX110-5L   | SMF 80D26L  | 260            | 173 | 222 | 12          | 56           | 70   | 120      | 600        | 730     | 0      | STD      | P/H    |
| SMF NX110-5    | SMF 80D26R  | 260            | 173 | 222 | 12          | 56           | 70   | 120      | 600        | 730     | 1      | STD      | P/H    |
| SMF N70L       | SMF 65D31L  | 303            | 173 | 225 | 12          | 56           | 70   | 105      | 560        | 690     | 0      | STD      | P/H    |
| SMF N70        | SMF 65D31R  | 303            | 173 | 225 | 12          | 56           | 70   | 105      | 560        | 690     | 1      | STD      | P/H    |
| SMF N70ZL      | SMF 75D31L  | 303            | 173 | 225 | 12          | 60           | 75   | 120      | 630        | 760     | 0      | STD      | P/H    |
| SMF N70Z       | SMF 75D31R  | 303            | 173 | 225 | 12          | 60           | 75   | 120      | 630        | 760     | 1      | STD      | P/H    |
| SMF NX120-7L   | SMF 95D31L  | 303            | 173 | 225 | 12          | 72           | 90   | 150      | 750        | 880     | 0      | STD      | P/H    |
| SMF NX120-7    | SMF 95D31R  | 303            | 173 | 225 | 12          | 72           | 90   | 150      | 750        | 880     | 1      | STD      | P/H    |

\*B3 BHD is available by customer's request

※ Specifications are subject to change without notice.

Sealed Maintenance Free Battery(ASIA)

Heavy-Duty Commercial Battery

| Old JIS    | New JIS    | Dimensions(mm) |     |     | Voltage (V) | Capacity(Ah) |      | RC (Min) | CCA (-18℃) | CA (0℃) | Layout | Terminal | Handle |
|------------|------------|----------------|-----|-----|-------------|--------------|------|----------|------------|---------|--------|----------|--------|
|            |            | L              | W   | TH  |             | 5HR          | 20HR |          |            |         |        |          |        |
| SMF 1000LA |            | 330            | 173 | 240 | 12          | 80           | 100  | 182      | 870        | 1000    | 0      | STD      | P/H    |
| SMF 1000RA |            | 330            | 173 | 240 | 12          | 80           | 100  | 182      | 870        | 1000    | 1      | STD      | P/H    |
| SMF 1000LS |            | 330            | 173 | 240 | 12          | 80           | 100  | 182      | 870        | 1000    | 0      | STD      | P/H    |
| SMF 1000RS |            | 330            | 173 | 240 | 12          | 80           | 100  | 182      | 870        | 1000    | 1      | STD      | P/H    |
| SMF N100L  | SMF 95E41L | 410            | 175 | 235 | 12          | 80           | 100  | 180      | 730        | 860     | 0      | STD      | R/N    |
| SMF N100   | SMF 95E41R | 410            | 175 | 235 | 12          | 80           | 100  | 180      | 730        | 860     | 1      | STD      | R/N    |
| SMF N120L  |            | 505            | 183 | 240 | 12          | 96           | 120  | 230      | 820        | 950     | 4      | STD      | R/N    |
| SMF N120   |            | 505            | 183 | 240 | 12          | 96           | 120  | 230      | 820        | 950     | 3      | STD      | R/N    |
| SMF N150L  |            | 508            | 222 | 241 | 12          | 120          | 150  | 300      | 1000       | 1130    | 4      | STD      | R/N    |
| SMF N150   |            | 508            | 222 | 241 | 12          | 120          | 150  | 300      | 1000       | 1130    | 3      | STD      | R/N    |
| SMF N200L  |            | 523            | 279 | 248 | 12          | 160          | 200  | 430      | 1150       | 1280    | 4      | STD      | R/N    |
| SMF N200   |            | 523            | 279 | 248 | 12          | 160          | 200  | 430      | 1150       | 1280    | 3      | STD      | R/N    |

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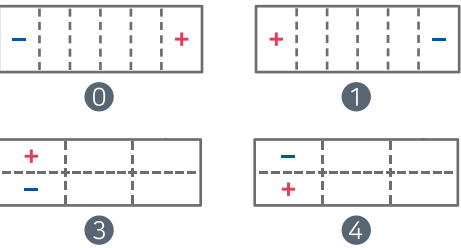
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DIN Type

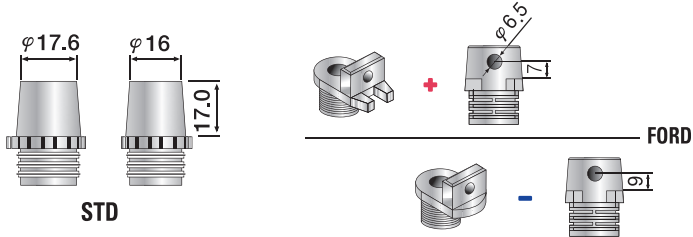
DEUTSCHES INSTITUT FUR NORMUNG TYPE CALCIUM MF BATTERY



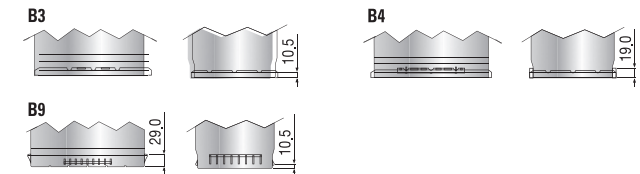
CELL LAYOUT



TERMINAL



BASE HOLD DOWN(BHD)



Sealed Maintenance Free Battery(Europe)

Passenger Car & Light Commercial Battery

| TYPE | Model     | Dimensions(mm) |     |     | Voltage<br>(V) | Capacity<br>20HR(AH) | RC<br>(Min) | CCA (-18°C) |      | Layout | Terminal | BHD   |
|------|-----------|----------------|-----|-----|----------------|----------------------|-------------|-------------|------|--------|----------|-------|
|      |           | L              | W   | TH  |                |                      |             | SAE         | EN   |        |          |       |
| LO   | SMF 54018 | 175            | 175 | 190 | 12             | 40                   | 70          | 350         | 330  | 0      | STD      | B3    |
| LB1  | SMF 53529 | 211            | 175 | 175 | 12             | 35                   | 52          | 340         | 340  | 0      | STD      | B3    |
|      | SMF 54316 | 211            | 175 | 175 | 12             | 43                   | 70          | 420         | 420  | 0      | STD      | B3    |
| L1   | SMF 54317 | 211            | 175 | 175 | 12             | 43                   | 70          | 420         | 420  | 0      | FORD     | B3    |
|      | SMF 54459 | 208            | 175 | 190 | 12             | 44                   | 71          | 350         | 330  | 0      | STD      | B3    |
| LB2  | SMF 54464 | 208            | 175 | 190 | 12             | 44                   | 71          | 350         | 330  | 1      | STD      | B3    |
|      | SMF 54519 | 245            | 175 | 175 | 12             | 45                   | 80          | 420         | 420  | 0      | STD      | B4    |
| L2   | SMF 55040 | 245            | 175 | 175 | 12             | 50                   | 86          | 460         | 460  | 0      | STD      | B3    |
|      | SMF 55457 | 245            | 175 | 175 | 12             | 54                   | 94          | 510         | 510  | 0      | STD      | B3    |
| LB3  | SMF 55458 | 245            | 175 | 175 | 12             | 54                   | 94          | 510         | 510  | 1      | STD      | B3    |
|      | SMF 55459 | 245            | 175 | 175 | 12             | 54                   | 94          | 510         | 510  | 0      | FORD     | B3    |
| L3   | SMF 56220 | 245            | 175 | 175 | 12             | 62                   | 100         | 540         | 540  | 0      | STD      | B3    |
|      | SMF 55559 | 245            | 175 | 190 | 12             | 55                   | 94          | 480         | 440  | 0      | STD      | B3    |
| LB4  | SMF 55565 | 245            | 175 | 190 | 12             | 55                   | 94          | 480         | 440  | 1      | STD      | B3    |
|      | SMF 56219 | 245            | 175 | 190 | 12             | 62                   | 100         | 510         | 480  | 0      | STD      | B3    |
| L4   | SMF 56217 | 245            | 175 | 190 | 12             | 62                   | 100         | 510         | 480  | 1      | STD      | B3    |
|      | SMF 56048 | 269            | 174 | 226 | 12             | 60                   | 100         | 500         | 460  | 0      | STD      | B9    |
| LB5  | SMF 56049 | 269            | 174 | 226 | 12             | 60                   | 100         | 500         | 460  | 1      | STD      | B9    |
|      | SMF 57029 | 269            | 174 | 226 | 12             | 70                   | 120         | 600         | 560  | 0      | STD      | B9    |
| L5   | SMF 57024 | 269            | 174 | 226 | 12             | 70                   | 120         | 600         | 560  | 1      | STD      | B9    |
|      | SMF 55414 | 277            | 175 | 175 | 12             | 54                   | 94          | 510         | 510  | 0      | STD      | B3/B4 |
| MAC  | SMF 55415 | 277            | 175 | 175 | 12             | 54                   | 94          | 510         | 510  | 1      | STD      | B3/B4 |
|      | SMF 56318 | 277            | 175 | 175 | 12             | 63                   | 110         | 580         | 580  | 0      | STD      | B4    |
| A    | SMF 56323 | 277            | 175 | 175 | 12             | 63                   | 110         | 580         | 580  | 0      | FORD     | B3    |
|      | SMF 56330 | 277            | 175 | 175 | 12             | 63                   | 110         | 580         | 580  | 0      | STD      | B3    |
| B    | SMF 57113 | 277            | 175 | 175 | 12             | 71                   | 115         | 620         | 620  | 0      | STD      | B3    |
|      | SMF 57114 | 277            | 175 | 175 | 12             | 71                   | 115         | 620         | 620  | 0      | FORD     | B3    |
| C    | SMF 56638 | 277            | 175 | 190 | 12             | 66                   | 110         | 580         | 540  | 0      | STD      | B3    |
|      | SMF 57220 | 277            | 175 | 190 | 12             | 72                   | 120         | 620         | 580  | 0      | STD      | B3    |
| LB4  | SMF 57219 | 277            | 175 | 190 | 12             | 72                   | 120         | 620         | 580  | 1      | STD      | B3    |
|      | SMF 57820 | 277            | 175 | 190 | 12             | 78                   | 140         | 700         | 660  | 0      | STD      | B3    |
| L4   | SMF 58014 | 315            | 175 | 175 | 12             | 80                   | 135         | 730         | 730  | 0      | STD      | B3    |
|      | SMF 59042 | 315            | 175 | 190 | 12             | 90                   | 160         | 780         | 720  | 0      | STD      | B3    |
| LB5  | SMF 59043 | 315            | 175 | 190 | 12             | 90                   | 160         | 780         | 720  | 1      | STD      | B3    |
|      | SMF 58515 | 353            | 175 | 175 | 12             | 85                   | 140         | 740         | 700  | 0      | STD      | B3    |
| L5   | SMF 59015 | 353            | 175 | 175 | 12             | 90                   | 156         | 820         | 820  | 0      | STD      | B3    |
|      | SMF 58827 | 353            | 175 | 190 | 12             | 88                   | 148         | 730         | 700  | 0      | STD      | B3    |
| MAC  | SMF 59218 | 353            | 175 | 190 | 12             | 92                   | 160         | 780         | 740  | 0      | STD      | B3    |
|      | SMF 60044 | 353            | 175 | 190 | 12             | 100                  | 170         | 850         | 820  | 0      | STD      | B3    |
| A    | SMF 61023 | 514            | 175 | 210 | 12             | 110                  | 215         | 800         | 800  | 3      | STD      | B3    |
|      | SMF 63541 | 514            | 175 | 210 | 12             | 135                  | 250         | 910         | 910  | 3      | STD      | B3    |
| B    | SMF 64020 | 513            | 189 | 223 | 12             | 140                  | 260         | 800         | 800  | 3      | STD      | B0/B3 |
|      | SMF 64028 | 513            | 189 | 223 | 12             | 140                  | 260         | 800         | 800  | 4      | STD      | B0/B3 |
| C    | SMF 67018 | 513            | 222 | 220 | 12             | 170                  | 330         | 900         | 900  | 3      | STD      | B0/B3 |
|      | SMF 67019 | 513            | 222 | 220 | 12             | 170                  | 330         | 900         | 900  | 4      | STD      | B0/B3 |
| LB4  | SMF 68032 | 513            | 222 | 220 | 12             | 180                  | 350         | 950         | 950  | 3      | STD      | B0/B3 |
|      | SMF 68033 | 513            | 222 | 220 | 12             | 180                  | 350         | 950         | 950  | 4      | STD      | B0/B3 |
| L5   | SMF 68021 | 518            | 273 | 241 | 12             | 180                  | 350         | 950         | 950  | 3      | STD      | B0    |
|      | SMF 70027 | 518            | 273 | 241 | 12             | 200                  | 390         | 1150        | 1150 | 3      | STD      | B0    |
| LB5  | SMF 73011 | 518            | 273 | 241 | 12             | 230                  | 430         | 1200        | 1200 | 3      | STD      | B0    |

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BCI Type

BATTERY COUNCIL INTERNATIONAL TYPE CALCIUM MF BATTERY



A. Passenger Car & Light Commercial Batteries

| Group | Model        | Dimensions(mm) |     |     | Voltage (V) | Capacity(Ah) |      | RC  | CCA (A) | CA (A) | Figure No. | Terminal |
|-------|--------------|----------------|-----|-----|-------------|--------------|------|-----|---------|--------|------------|----------|
|       |              | L              | W   | TH  |             | 5HR          | 20HR |     |         |        |            |          |
| 58    | SMF 58-450   | 236            | 183 | 177 | 12          | 40           | 50   | 74  | 450     | 560    | 21         | STD      |
|       | SMF 58-530   | 236            | 183 | 177 | 12          | 48           | 60   | 86  | 530     | 660    | 21         | STD      |
| 65    | SMF 65-660   | 305            | 190 | 191 | 12          | 58           | 72   | 120 | 660     | 780    | 21         | STD      |
|       | SMF 65-700   | 305            | 190 | 191 | 12          | 62           | 78   | 130 | 700     | 830    | 21         | STD      |
|       | SMF 65-820   | 305            | 190 | 191 | 12          | 70           | 88   | 140 | 820     | 930    | 21         | STD      |
| 22F   | SMF 22F-470  | 240            | 174 | 205 | 12          | 40           | 50   | 86  | 470     | 590    | 11F        | STD/DUAL |
|       | SMF 22F-530  | 240            | 174 | 205 | 12          | 43           | 54   | 95  | 530     | 650    | 11F        | STD/DUAL |
| 26    | SMF 26-420   | 208            | 171 | 205 | 12          | 34           | 42   | 60  | 420     | 550    | 10         | STD      |
|       | SMF 26-560   | 208            | 171 | 205 | 12          | 43           | 54   | 86  | 560     | 690    | 10         | STD      |
| 70    | SMF 70-420   | 208            | 179 | 185 | 12          | 34           | 42   | 60  | 420     | 550    | 17         | SIDE     |
|       | SMF 70-560   | 208            | 179 | 185 | 12          | 43           | 54   | 86  | 560     | 690    | 17         | SIDE     |
| 70DT  | SMF 70DT-420 | 208            | 179 | 205 | 12          | 34           | 42   | 60  | 420     | 550    | 10,17      | DUAL     |
|       | SMF 70DT-560 | 208            | 179 | 205 | 12          | 43           | 54   | 86  | 560     | 690    | 10,17      | DUAL     |
| 86    | SMF 86-520   | 231            | 171 | 205 | 12          | 40           | 50   | 79  | 520     | 650    | 10         | STD      |
|       | SMF 86-710   | 231            | 171 | 205 | 12          | 53           | 66   | 110 | 710     | 840    | 10         | STD      |
| 75    | SMF 75-520   | 231            | 179 | 185 | 12          | 40           | 50   | 79  | 520     | 650    | 17         | SIDE     |
|       | SMF 75-710   | 231            | 179 | 185 | 12          | 53           | 66   | 110 | 710     | 840    | 17         | SIDE     |
| 75DT  | SMF 75DT-520 | 231            | 179 | 205 | 12          | 40           | 50   | 78  | 520     | 650    | 10,17      | DUAL     |
|       | SMF 75DT-710 | 231            | 179 | 205 | 12          | 53           | 66   | 110 | 710     | 840    | 10,17      | DUAL     |
| 34    | SMF 34-615   | 260            | 171 | 205 | 12          | 46           | 58   | 94  | 615     | 745    | 10         | STD      |
|       | SMF 34-780   | 260            | 171 | 205 | 12          | 59           | 74   | 124 | 780     | 910    | 10         | STD      |
| 78    | SMF 78-615   | 260            | 179 | 185 | 12          | 46           | 58   | 94  | 615     | 745    | 17         | SIDE     |
|       | SMF 78-780   | 260            | 179 | 185 | 12          | 59           | 74   | 124 | 780     | 910    | 17         | SIDE     |
| 78DT  | SMF 78DT-615 | 260            | 179 | 205 | 12          | 46           | 58   | 94  | 615     | 745    | 10,17      | DUAL     |
|       | SMF 78DT-780 | 260            | 179 | 205 | 12          | 59           | 74   | 124 | 780     | 910    | 10,17      | DUAL     |

\*More or less CCA Batteries are available up on customer's request

※ Specifications are subject to change without notice.

B. Heavy-Duty Commercial Batteries

| Group | Model        | Dimensions(mm) |     |     | Voltage (V) | Capacity(Ah) |      | RC  | CCA (A) | CA (A) | Figure No. | Terminal |
|-------|--------------|----------------|-----|-----|-------------|--------------|------|-----|---------|--------|------------|----------|
|       |              | L              | W   | TH  |             | 5HR          | 20HR |     |         |        |            |          |
| 31S   | SMF 31-730S  | 330            | 173 | 240 | 12          | 68           | 85   | 150 | 730     | 860    | 18         | STUD     |
|       | SMF 31-1000S | 330            | 173 | 240 | 12          | 96           | 120  | 205 | 1000    | 1150   | 18         | STUD     |
| 31A   | SMF 31-730A  | 330            | 173 | 240 | 12          | 68           | 85   | 150 | 730     | 860    | 18         | STD      |
|       | SMF 31-1000A | 330            | 173 | 240 | 12          | 96           | 120  | 205 | 1000    | 1150   | 18         | STD      |
| 30H   | SMF 30H-830  | 330            | 173 | 229 | 12          | 80           | 100  | 180 | 830     | 960    | 10         | STD      |
|       | SMF 30HL-830 | 330            | 173 | 229 | 12          | 80           | 100  | 180 | 830     | 960    | 10         | STD      |
| 4D    | SMF 4D-850   | 508            | 222 | 241 | 12          | 112          | 140  | 270 | 850     | 980    | 8          | STD      |
|       | SMF 4D-1000  | 508            | 222 | 241 | 12          | 120          | 150  | 300 | 1000    | 1080   | 8          | STD      |
| 8D    | SMF 8D-1200  | 523            | 279 | 250 | 12          | 144          | 180  | 315 | 1200    | 1330   | 8          | STD      |
|       | SMF 8D-1400  | 523            | 279 | 250 | 12          | 160          | 200  | 400 | 1400    | 1530   | 8          | STD      |

\*More or less CCA Batteries are available up on customer's request

※ Specifications are subject to change without notice.

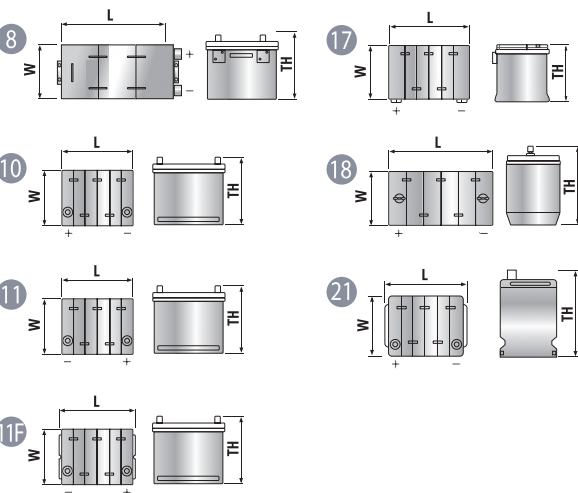
C. General Utility & Marine Batteries

| Group | Model         | Dimensions(mm) |     |     | Voltage (V) | Capacity(Ah) |      | RC  | CCA (A) | CA (A) | Figure No. | Terminal |
|-------|---------------|----------------|-----|-----|-------------|--------------|------|-----|---------|--------|------------|----------|
|       |               | L              | W   | TH  |             | 5HR          | 20HR |     |         |        |            |          |
| U1    | SMF U1-250    | 195            | 130 | 183 | 12          | 18           | 23   | 35  | 250     | 360    | 10         | LUG      |
|       | SMF U1-330    | 195            | 130 | 183 | 12          | 24           | 30   | 45  | 330     | 440    | 10         | LUG      |
| U1R   | SMF U1R-250   | 195            | 130 | 183 | 12          | 18           | 23   | 35  | 250     | 360    | 11         | LUG      |
|       | SMF U1R-330   | 195            | 130 | 183 | 12          | 24           | 30   | 45  | 330     | 440    | 11         | LUG      |
| M24   | SMF 24MS-500  | 260            | 174 | 222 | 12          | 52           | 65   | 85  | 500     | 630    | 10         | TM       |
|       | SMF 24MS-680  | 260            | 174 | 222 | 12          | 64           | 80   | 115 | 680     | 810    | 10         | TM       |
| M27   | SMF 27MS-730  | 304            | 173 | 225 | 12          | 72           | 90   | 125 | 730     | 860    | 10         | TM       |
|       | SMF 27MS-780  | 304            | 173 | 225 | 12          | 76           | 95   | 130 | 780     | 910    | 10         | TM       |
| M31   | SMF 31MS-850  | 330            | 173 | 240 | 12          | 80           | 100  | 180 | 850     | 980    | 18         | TM       |
|       | SMF 24HCM-500 | 260            | 174 | 222 | 12          | 56           | 70   | 130 | 500     | 630    | 10         | TM       |
| 24HCM | SMF 24HCM-600 | 260            | 174 | 222 | 12          | 66           | 82   | 140 | 600     | 730    | 10         | TM       |
|       | SMF 27HCM-660 | 304            | 173 | 225 | 12          | 70           | 88   | 160 | 660     | 790    | 10         | TM       |
| 27HCM | SMF 27HCM-700 | 304            | 173 | 225 | 12          | 76           | 95   | 170 | 700     | 830    | 10         | TM       |
|       | SMF 31HC-780  | 330            | 173 | 240 | 12          | 80           | 100  | 180 | 700     | 830    | 18         | STD/STUD |

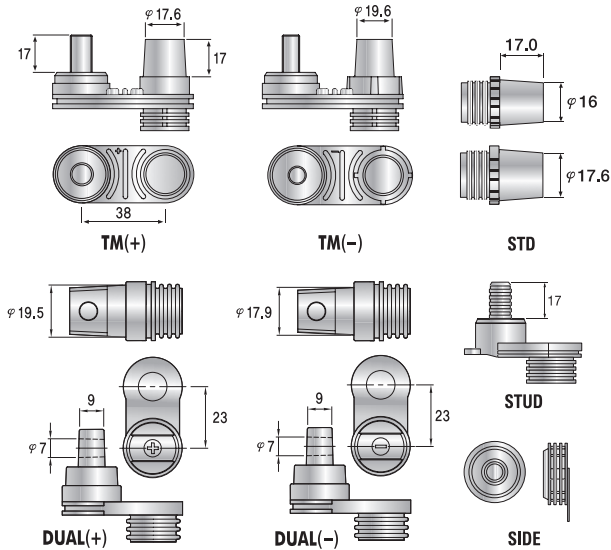
\*More or less CCA Batteries are available up on customer's request

※ Specifications are subject to change without notice.

BCI FIGURE



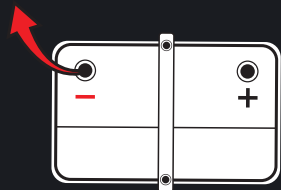
TERMINAL



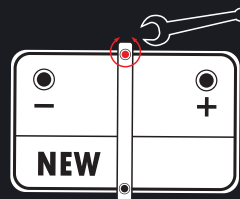
## PROCEDURE REPLACEMENT



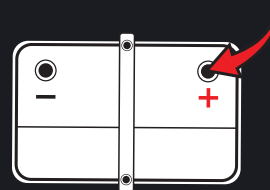
1 Switch off the engine and keep sparks, flames, cigarettes away from battery at all times, before removing the old battery.



2 Disconnect the negative terminal, first and avoid short-circuit caused by tools, when removing old battery.



3 Clean corroded parts in tray and cable clamps, before installation of new battery.



4 Connect the positive terminal, first and check the terminal clamps for tight fit, when installing new battery.

## CAUTION

- No Sparks, Flames, or Smoking.
- Charge in well ventilated area. Remove from vehicle prior to charging.
- Read your vehicle, booster, and battery manuals prior to charging.
- In case of exposure to acid, flush immediately with water. Then get immediate medical help.
- Do not block or alter vent hole.
- Keep out of reach of children.

**WARNING!** LEAD ACID BATTERIES GENERATE EXPLOSIVE GASES. KEEP SPARKS, FLAMES, & LIGHTED CIGARETTES AWAY FROM BATTERY, WHEN CHARGING OR USING BATTERY IN AN ENCLOSED SPACE, PROVIDE VENTILATION. THIS BATTERY CONTAINS SULPHURIC ACID. AVOID CONTACT WITH SKIN, EYES, OR CLOTHING. IN EVENT OF ACCIDENT FLUSH WITH WATER AND CALL A PHYSICIAN IMMEDIATELY

**MAXIMUM RECHARGING VOLTAGE IS 14.6V**



BATTERY, WET,  
FILLED WITH ACID



Flammables



Read Instructions  
Carefully



Keep out of  
Reach of Children



Cautions of  
Sulfuric Acid



Shield Eyes



Explosive